

Supplementary Material

Constraint-Driven Sparse Fragment Reconstruction for Digital Forensics

This supplement holds evidence that supports the main paper without carrying a headline claim. Every table is generated from the released CSVs by `scripts/render_stats_tables.py` and `scripts/update_paper_tables.py`, the same generators that populate the main paper, so the numbers here cannot drift from the artefacts they come from.

1 Constraint ablation

The contribution of each constraint family, averaged over all 20 cells.

Table S1: Constraint ablation, averaged over the full 20-cell sweep and seeds $\{0, 1, 2\}$. Every variant retains the C1 data term, the ℓ_1 sparsity prior, and the final C1 projection (so C1 = 0 throughout, omitted). CVR columns in %.

Variant	PSNR (dB)	SSIM	HRP	CVR C2	CVR C3	CVR C4
C1 only	24.41	0.716	1.76	1.55	5.18	2.69
C1+C2 (TV)	25.56	0.752	1.76	1.59	0.08	2.08
C1+C3 (bounds)	25.09	0.727	1.58	1.52	0.00	2.66
C1+C4 (freq.)	24.07	0.725	1.83	1.68	5.38	1.69
Full CSFR	25.68	0.757	1.73	1.46	0.00	1.85

2 Runtime and memory

Wall-clock and peak memory per patch, measured on the CPU reference machine.

3 Iteration-budget sensitivity

CSFR PSNR against solver budget on a 100-patch, seed-0 subset. The gap column compares against the strongest baseline’s full-cell value, so it slightly understates CSFR where the subset is easier than the full cell.

4 Bounded-Unknown and the signal-covering subset

Abstention rates and the continuous isolated-missing-fraction per cell. The binary flag saturates by family and level, which is why the continuous statistic is the one reported in the main paper.

Table S2: Per-patch runtime and peak memory, CPU, 64×64 patches. ms/patch and per-batch mean $\pm$ SD over five repeats of a 32-patch batch; peak is `tracemalloc` peak allocation. Full environment in `results/bench/environment.json`.

Method	ms/patch	mean (s)	SD (s)	median (s)	peak (MB)
Zero-fill	0.03	0.001	0.000	0.001	1.6
Linear interpolation	84.64	2.708	0.290	2.596	2.4
Inpainting (Telea)	1.61	0.052	0.006	0.049	1.5
Inpainting (Navier–Stokes)	1.18	0.038	0.001	0.038	1.5
Dictionary learning	1.46	0.047	0.001	0.047	1.5
CSFR (proposed)	44.35	1.419	0.105	1.435	1.0

Table S3: Iteration-budget sensitivity. CSFR PSNR (dB) is recomputed on a 100-patch, seed-0 subset at each budget; the “Gap@4800” column is CSFR at $T=4800$ minus the strongest baseline’s full-cell (300-patch, seed-0) PSNR in that cell, so it slightly understates CSFR where the subset is easier than the full cell. Extending the budget closes most of the CF3/CF4 deficit and reverses the CF2 deficit, establishing that the reported 600-iteration losses on these families are substantially an under-convergence effect.

Cell	T=600	T=1200	T=2400	T=4800	Gap@4800
CF1 L5	28.15	28.15	28.15	28.15	+0.10
CF2 L3	25.78	27.81	28.00	28.00	+1.46
CF2 L5	16.57	20.44	22.32	22.45	+1.32
CF3 L5	11.21	12.36	14.56	17.44	-0.61
CF4 L5	11.03	12.14	14.39	17.62	-0.61

Table S4: Bounded-Unknown and abstention rates by cell. The binary Bounded-Unknown flag at threshold 0.5 on isolated-missing-fraction saturates: it is zero for every CF1 cell and for CF3/CF4 L1, unity for CF3/CF4 from L2 upward and CF2 from L3 upward, and varies within a cell only at CF2 L1 and L2. It is a family-and-level indicator rather than a per-patch signal. The continuous isolated-missing-fraction (columns: mean, SD, median) is reported beside it and should be preferred for analysis. The signal-covering subset count n_{sig} is the number of patches in which at least a quarter of the centre box is erased, so that the corruption overlaps the digit; it is defined on the corruption geometry, not on constraint violations. It is empty for CF1 at L1–L2 and for CF3/CF4 at L1–L3, and saturates to 1000 for every family from L4 upward, leaving CF1 L3 and CF2 L1–L3 as the only cells with within-cell variation.

Fam.	Lvl	BU rate	IMF mean	IMF SD	IMF median	n_{sig}
CF1	L1	0.000	0.000	0.001	0.000	0
CF1	L2	0.000	0.001	0.003	0.000	0
CF1	L3	0.000	0.010	0.006	0.010	982
CF1	L4	0.000	0.070	0.012	0.070	1000
CF1	L5	0.000	0.253	0.016	0.252	1000
CF2	L1	0.880	0.612	0.091	0.619	352
CF2	L2	0.992	0.662	0.067	0.664	760
CF2	L3	1.000	0.710	0.050	0.713	991
CF2	L4	1.000	0.778	0.036	0.781	1000
CF2	L5	1.000	0.849	0.022	0.851	1000
CF3	L1	0.000	0.500	0.000	0.500	0
CF3	L2	1.000	0.800	0.000	0.800	0
CF3	L3	1.000	0.900	0.000	0.900	0
CF3	L4	1.000	0.938	0.000	0.938	1000
CF3	L5	1.000	0.955	0.000	0.955	1000
CF4	L1	0.000	0.500	0.000	0.500	0
CF4	L2	1.000	0.800	0.000	0.800	0
CF4	L3	1.000	0.900	0.000	0.900	0
CF4	L4	1.000	0.938	0.000	0.938	1000
CF4	L5	1.000	0.955	0.000	0.955	1000